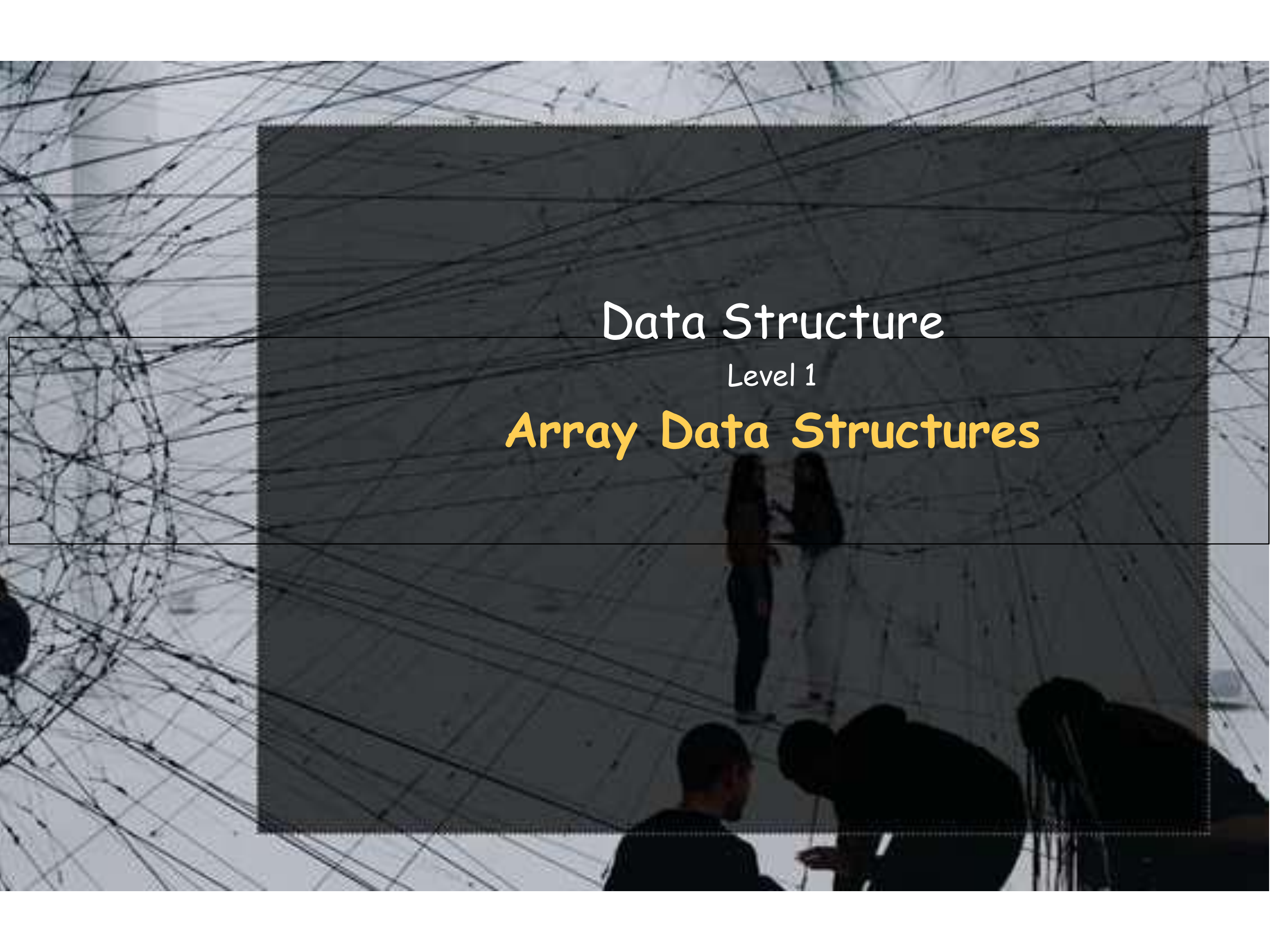


Data Structure Level 1

Array Data Structures

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, which adds a human element to the technical theme.

Data Structure

Level 1

Array Data Structures

Array Data Structure:

- Array: is a variable that can store multiple values of the same type.
- Array: An array is a collection of data items stored at contiguous memory locations.
- The idea is to store multiple items of the (homogeneous - same type) together

Array Data Structure:

```
int x[5] = { 22, 18, 2, 55, 520 };
```

Array
Members



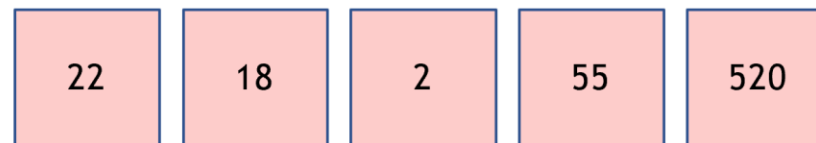
x[0]

x[1]

x[2]

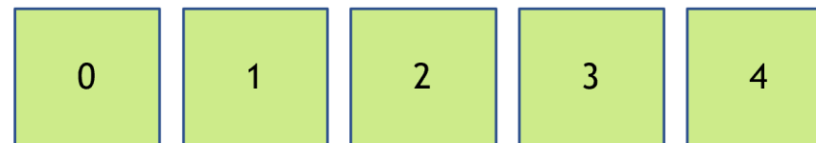
x[3]

x[4]

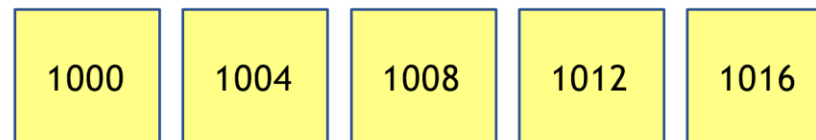


Array Values

Index starts
from zero

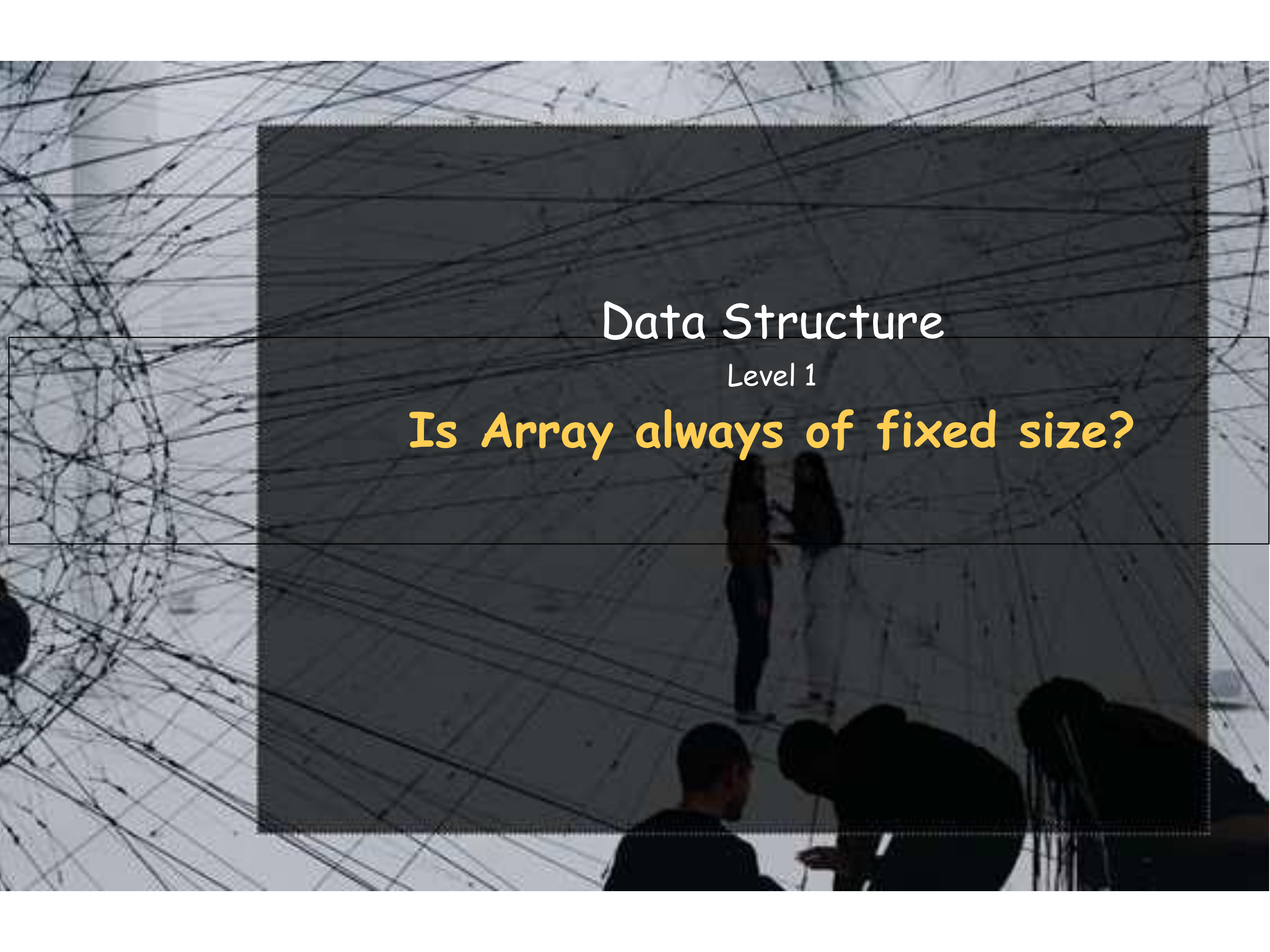


Array Indexes



Memory addresses

Memory

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Data Structure

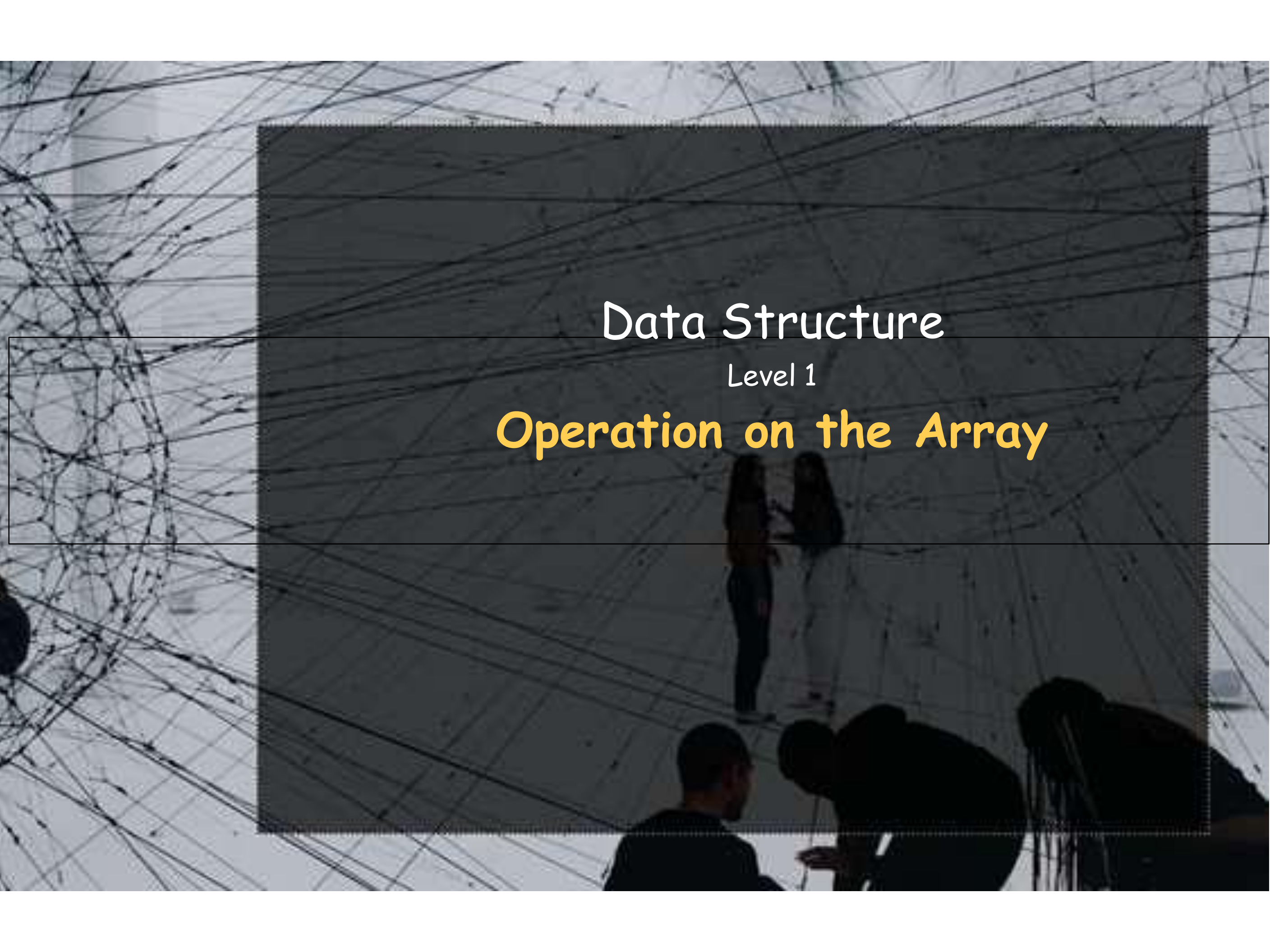
Level 1

Is Array always of fixed size?

Is Array always of fixed size?

Depends on the language for example: In C language, the array has a fixed size meaning once the size is given to it, it cannot be changed i.e. you can't shrink it nor can you expand it.

In C++ you can have dynamic arrays.

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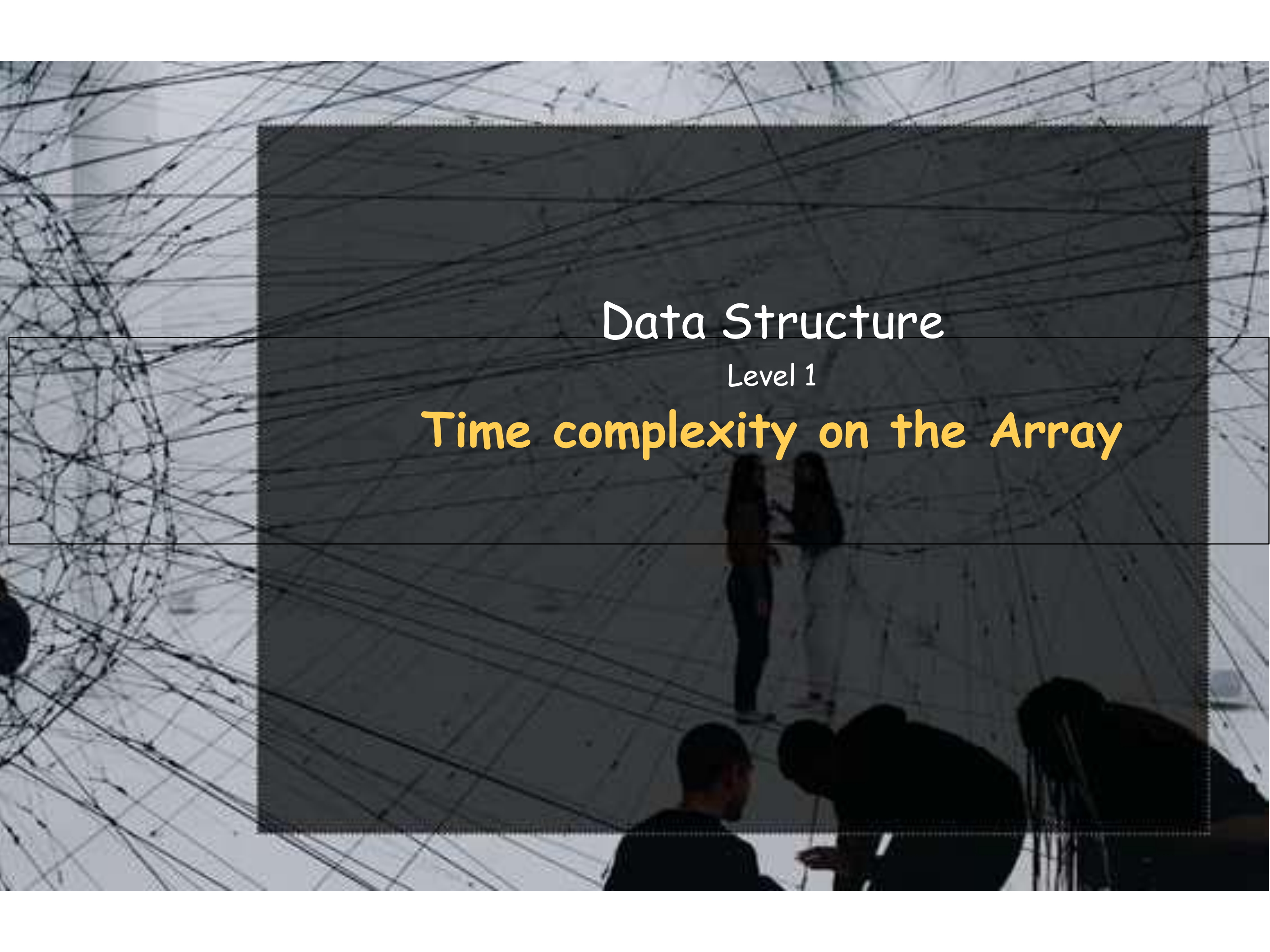
Data Structure

Level 1

Operation on the Array

Operation on the Array

- Arrays allow random access to elements. This makes accessing elements by position faster.
- Hence operation like searching, insertion, and access becomes really efficient.
- Array elements can be accessed using the loops.

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Data Structure

Level 1

Time complexity on the Array

Time complexity on the Array

- Insertion: We try to insert a value to a particular array index position, as the array provides random access it can be done easily using the assignment operator.

💡 `arr[2] = 10;`

- Time Complexity:
 - $O(1)$ to insert a single element
 - $O(N)$ to insert all the array elements [where N is the size of the array]

Time complexity on the Array

■ Access elements in Array: Accessing array elements become extremely important, in order to perform operations on arrays.

💡 `return arr[2];`

■ Time Complexity:

$O(1)$ to insert a single element

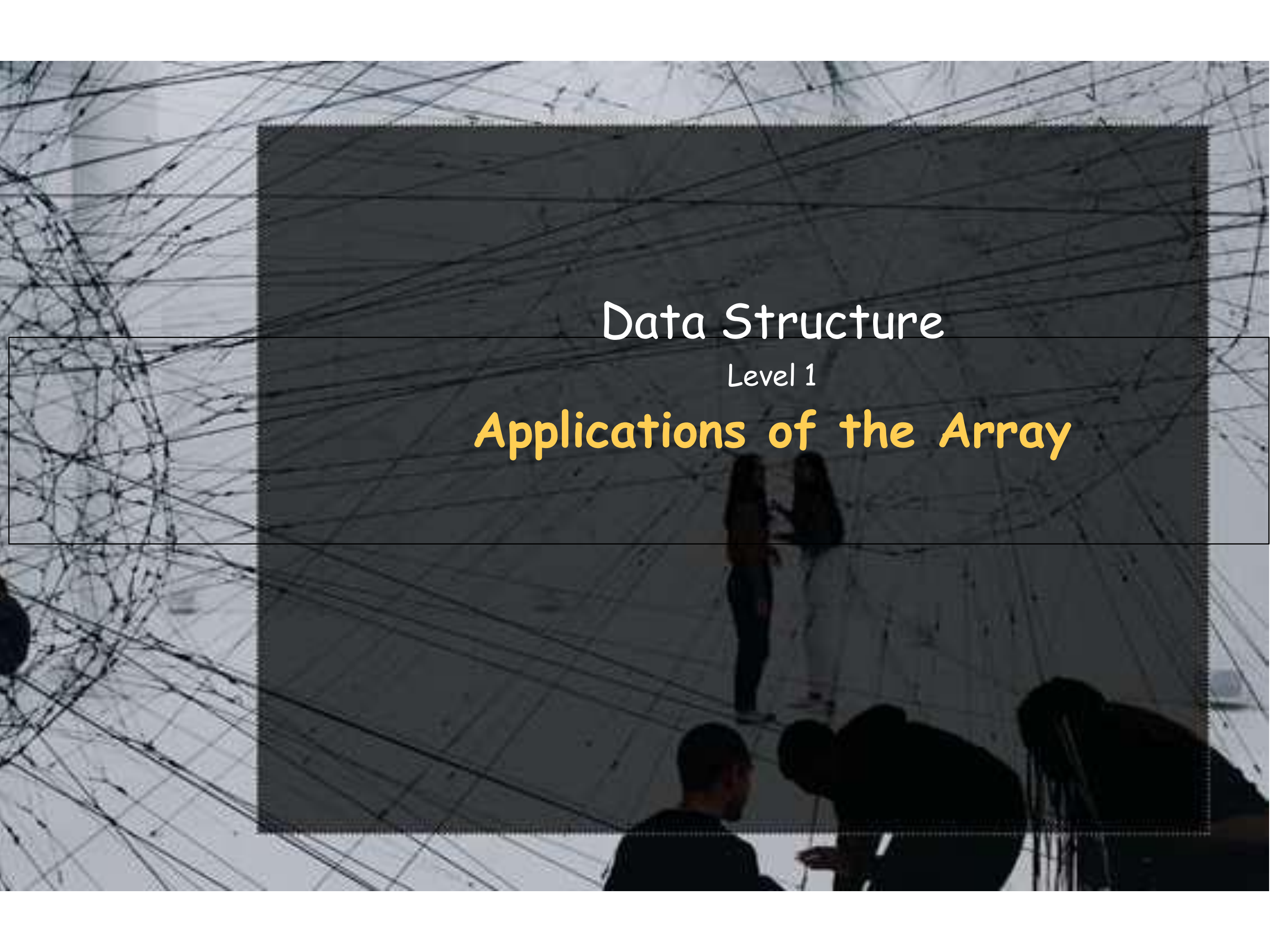
$O(N)$ to insert all the array elements [where N is the size of the array]

Time complexity on the Array

- Searching in Array: We try to find a particular value in the array, in order to do that we need to access all the array elements and look for the particular value.

```
// searching for value 55 in the array;  
Loop from i = 0 to 10:  
    check if arr[i] = 55:  
        return true;
```

- Time Complexity:
 $O(N)$ [where N is the size of the array]

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a web or a neural network. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Data Structure

Level 1

Applications of the Array

Applications of the Array

- Array stores data elements of the same data type.
- Arrays are used when the size of the data set is known.
- Used in solving matrix problems.
- Applied as a lookup table in computer.
- Databases records are also implemented by the array.
- Helps in implementing sorting algorithm.
- The different variables of the same type can be saved under one name.
- Arrays can be used for CPU scheduling.
- Used to Implement other data structures like Stacks, Queues, Heaps, Hash tables, etc.

Thank you 😊